MUC02 — Credit Pre-Screening Engine

Team Section Ownership & Individual Reflections

Loan Intake Optimisation | KL University (KLU)

1. Team Section Ownership

Team Member	Section / Responsibility	Notebook Steps
Chetan Kalagarla	Data Preparation & Quality Analysis	Step 5.1
Bezavada Raga Susmitha	Approval Drivers, Affordability & Fairness Analysis	Steps 5.2, 5.3, 5.6
Bandi Ashok	Model Building & Threshold Selection	Steps 5.4, 5.5
Chindanur Nakul Sriraj	Policy Briefing, Limitations & Documentation	Step 5.7

2. Individual Reflections

Chetan Kalagarla

Data Preparation & Quality Analysis

My part of the project was centred on preparing the loan application data and checking its quality before any analysis began. I went through the structure of the dataset, which has 12,000 applications and 19 columns, and checked the data types, missing values and duplicate rows. No duplicates were found, but 441 monthly income values were missing and 1,039 credit scores were blank. Missing income was filled using the median for each employment type, and 12 unusually high salaried incomes were corrected by dividing them by 12. The blank credit scores were not removed: I created a new-to-credit flag first, so these applicants stayed in the analysis, and all 12,000 rows were retained. This section taught me that careful, documented cleaning is what makes later results trustworthy. Watching the teammates build the approval analysis and the model on these prepared fields showed me how directly data quality affects the final business conclusions.

Bezavada Raga Susmitha

Approval Drivers, Affordability & Fairness Analysis

My contribution centred on understanding what drives historical approvals and whether the pattern looked fair. I compared approval rates across credit-score bands and employment types, and then studied affordability through FOIR. Approval was only 2.0% for scores below 600 but reached 85.2% at 800 and above, and it fell from 65.5% for FOIR below 30% to 15.6% above 55%. For the fairness check, gender was not used as a model feature, but I compared approval rates by gender (50.4% female and 54.1% male) and noticed that the gap was smaller within some employment types, while women had a lower median income (INR 45,000 against INR 55,000) and a higher Gig / Contract share (21.4% against 11.6%). At the 0.90 threshold, the auto-decline rate was 15.6% for women and 15.0% for men. This is a first check, not proof of fairness. From the data preparation work I learned why the new-to-credit flag mattered, and it made me more careful to describe these results as patterns in past decisions, not causes.

Bandi Ashok

Model Building & Threshold Selection

My focus was on building the logistic-regression pre-screener and choosing the auto-decline threshold. The model predicts a historical decline, and we kept identifiers, city, gender and the decision itself out of the features. On the test set it reached 79.1% accuracy, 79.1% precision, 75.5% recall and a ROC-AUC of 0.867. The key lesson was that accuracy alone is not enough. At the default 0.50 cut-off, 20.9% of auto-declines had been historically approved, which is far above the bank's 5% limit. By comparing thresholds, I found that 0.90 was the lowest tested value within the limit, with 15.2% of applications auto-declined and 4.7% wrongly declined. Using the prepared data and the affordability features from my teammates made the model stronger, and I learned how to turn model outputs into a decision rule a business can actually use.

Chindanur Nakul Sriraj

Policy Briefing, Limitations & Documentation

My responsibility was to bring the results together in the final credit policy briefing and to keep the project documentation clear. I worked from the outputs of every section to state the recommendation: a 0.90 decline-probability threshold, which would auto-decline about 456 of 3,000 monthly applications and free an estimated 190.1 analyst hours. I also made sure the limitations were stated plainly. The model learns the historical credit team's decisions, not actual repayment or default, its test results do not guarantee future outcomes, and a group-level fairness check is only a first diagnostic. I also organised the notebook, dataset and summary so that a reader unfamiliar with the project can follow the objective, the workflow and the results. Going through each section showed me that good documentation explains not just what was done, but why, and what must be kept in mind before the results are used in real lending decisions.

Prepared for MUC02 — Credit Pre-Screening Engine. Figures quoted above are taken from the project notebook (12,000 applications, held-out test-set results).